

# Fraud Detection & Claims Performance



Total Claims  
**1,000**



Fraud Cases  
**247**

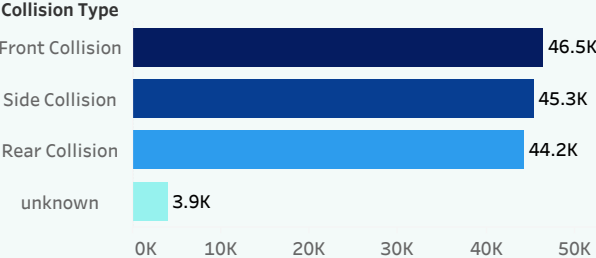


Avg Policy Premium  
**\$1,256**

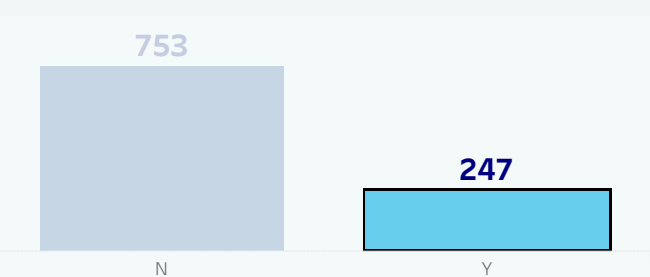


Fraud Rate  
**24.7%**

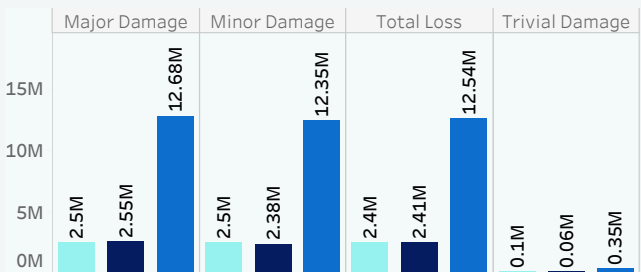
## Avg Vehicle Claim by Collision Type



## Fraud vs Non-Fraud Distribution



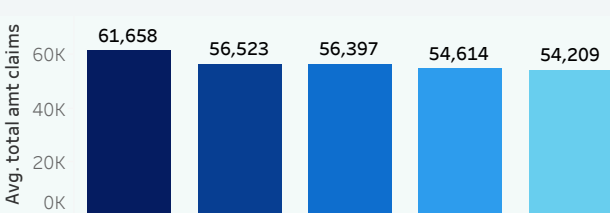
## Claims by incident severity



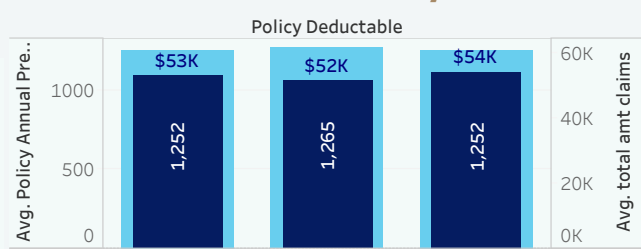
## Claims Volume & Fraud Rate by Hour of Day



## Top 5 Occupations by Average Claim Amount



## Premium & Total Claims by deductible



# Fraud Risk & Model Performance Dashboard



Accuracy  
41%



Sensitivity  
97.96%



Specificity  
22.52%



Fraud Rate  
24.50%

Threshold<sup>0.2</sup>

## Model Confusion Matrix

Predicted label

No

Yes

actu..

No

34  
TN

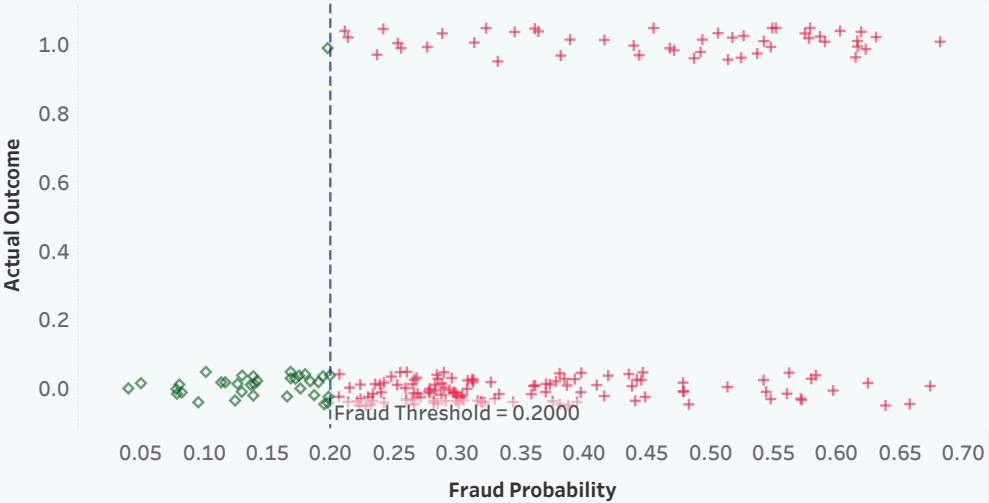
117  
FP

Yes

1  
FN

48  
TP

## Fraud Probability vs Actual Outcome



**High Fraud Detection** : The model captures ~98% of fraud cases (very high sensitivity) with almost no misses, ensuring strong protection against fraud losses.